Liquid Turbine Flowmeter Communication Protocol

MODBUS- RTU

Modbus Poll software RTU connection settings:

Display Option – Floating Pt (all data displayed as floating-point numbers)

Function 03: Read Holding Registers (4X)

Device id: internal address of the instrument

Address: starting address of instrument parameters, from 1 to 14

Length: number of registers; Length + Address ≤ 14

Parameter addresses:

40001-2: Temperature (for the liquid turbine, this reading is always 0)

40003-4: Instantaneous flow

40005-6: Flow velocity

40007-8: Frequency

40009-10: Cumulative flow – hundreds digit and above (example: if the value above hundreds is 1234)

40011-12: Cumulative flow – below the hundreds digit (e.g., below hundreds is 87.89)

Cumulative flow = 1234 × 100 + 87.89 = 123487.89

40013-14: Current instantaneous flow unit (0: m^3/h, 1: L/m, 2: kg/h, 3: L/h, 4: T/h, 5: kg/m, 6: m^3/m)

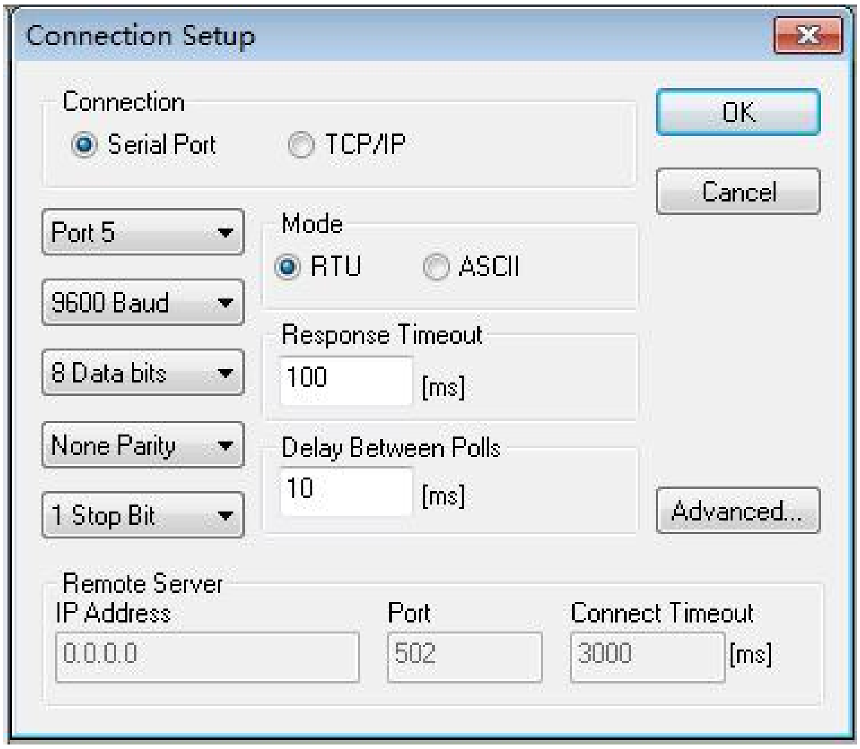
Note: This parameter address has an offset added by default – the actual parameter address is (address shown) minus 1.

(Clear command) Function 04: Write command

Parameter address: 30001

Write 99 to 30001. Data format: unsigned integer.

Modbus Poll operation interface:



Software settings: 9600 baud, 8 data bits, 1 stop bit, no parity.

	Alias	00000	Alias	00010
0		0.000000		0.875651
1				
2		0.000000		0.000000
3				
4		0.000000		
5				
6		0.000000		
7				
8		0.000000		
9				

Example:

Master request: 01 03 00 00 00 0E C4 0E (hexadecimal)

01 – device address

03 – function code

00 00 – starting register address

00 0E – number of registers (14)

C4 0E – CRC check

Slave response frame:

01 03 1C 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2A A7

3F 60 00 00 00 00 8D 0F

01 – device address

03 – function code

1C – byte count (0E registers = 14 registers × 2 bytes = 28 bytes)

The bytes 2A A7 3F 60 are the 11th and 12th registers (representing cumulative flow below the hundreds digit). These bytes need to be swapped and reordered before parsing (order: 3412).

That is, 3F 60 2A A7 – after parsing, the value is the same as read by the software: 0.87565.

Data type: All parsed values use the same data type (floating-point, byte order 3412).